



BITS Pilani
Pilani Campus



Lecture-1

Introduction

04-08-2026



Process Design..

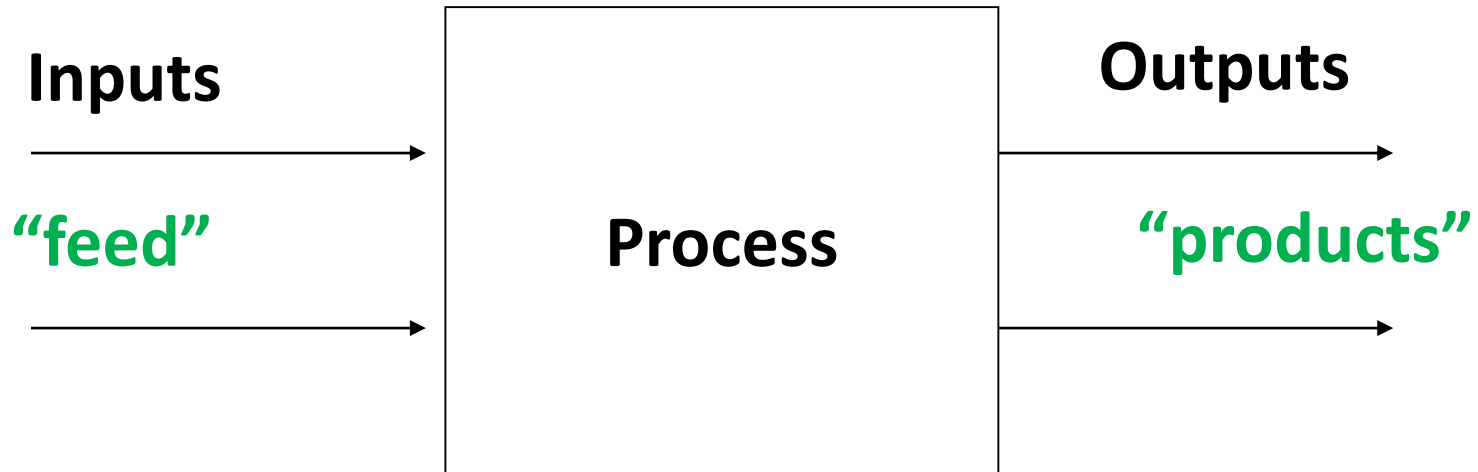
"It is a combination of science and art in a creative activity that helps to make process design such a fascinating challenge to an engineer..."



Process Design



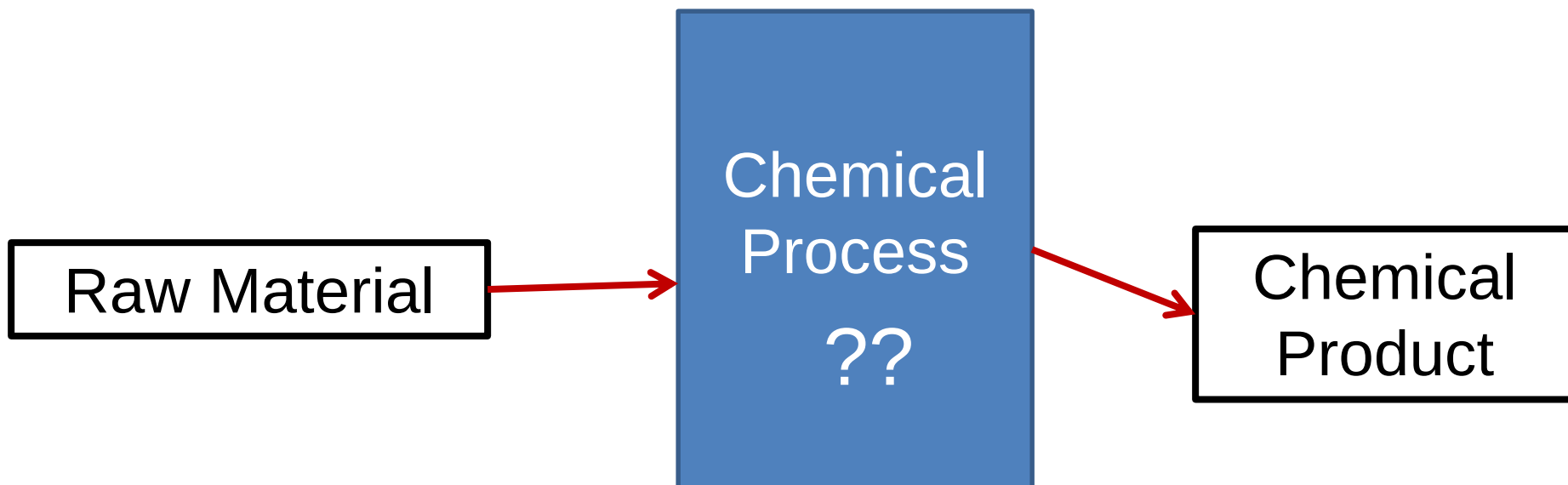
PROCESS





DESIGN

Definition of Chemical Process Design



Chemical process design is about finding a sustainable process that can convert the raw materials to the desired chemical products



Creative Aspects of Process Design



- The Purpose of Engineering
 - to create new material wealth
- This goal in Chemical Engineering is accomplished
 - via the chemical transformation
 - and/or separation of materials

Creative Aspects of Process Design



- 50% of the chemical products sold
 - were developed during the last few decades (30 – 40 years)
 - Indication of the tremendous success of engineering effort
- Process and Plant Design
 - Creative activity whereby
 - Generate ideas, and translate them into equations and processes for producing new materials
 - or for significantly upgrading the value of existing materials

Your role in the chemical process



Process synthesis vs. Process analysis



Strategy for Process Synthesis and Analysis

- The goal of a conceptual design is:
 - To find the best process flow sheet
 - To estimate the optimum design conditions
- There can be many process alternatives to be considered
- There are many possibilities to consider with only a small chance of success
 - 10^4 – 10^9 alternatives can be generated for a single product plant (since design problems are under-defined)



Contd..

- In some cases it is possible to use design guidelines (**rules of thumb or heuristics**)
 - to make some decisions about the structure of the flow sheet and/or
 - to set the values of some of the design variables
- In the absence of heuristics - Use shortcut design methods



Problem Areas Synthesis and Analysis

- Design problems are **underdefined**
- To supply this missing information, we must make assumptions about
 - What type of process units should be used?
 - How are they interconnected?
 - What temperatures, pressures, flow rates are required?

... **Synthesis Activity**



Contd..

- Synthesis is difficult because there are very large number ($10^4 - 10^9$) of ways to accomplish same goal
- Hence design problems are very open-ended



Objective

- We have to find the process alternative (out of $10^4 - 10^9$) possibilities
 - That has the lowest cost
 - Process is safe
 - Satisfy environmental constraints
 - Easy to start up and operate etc.

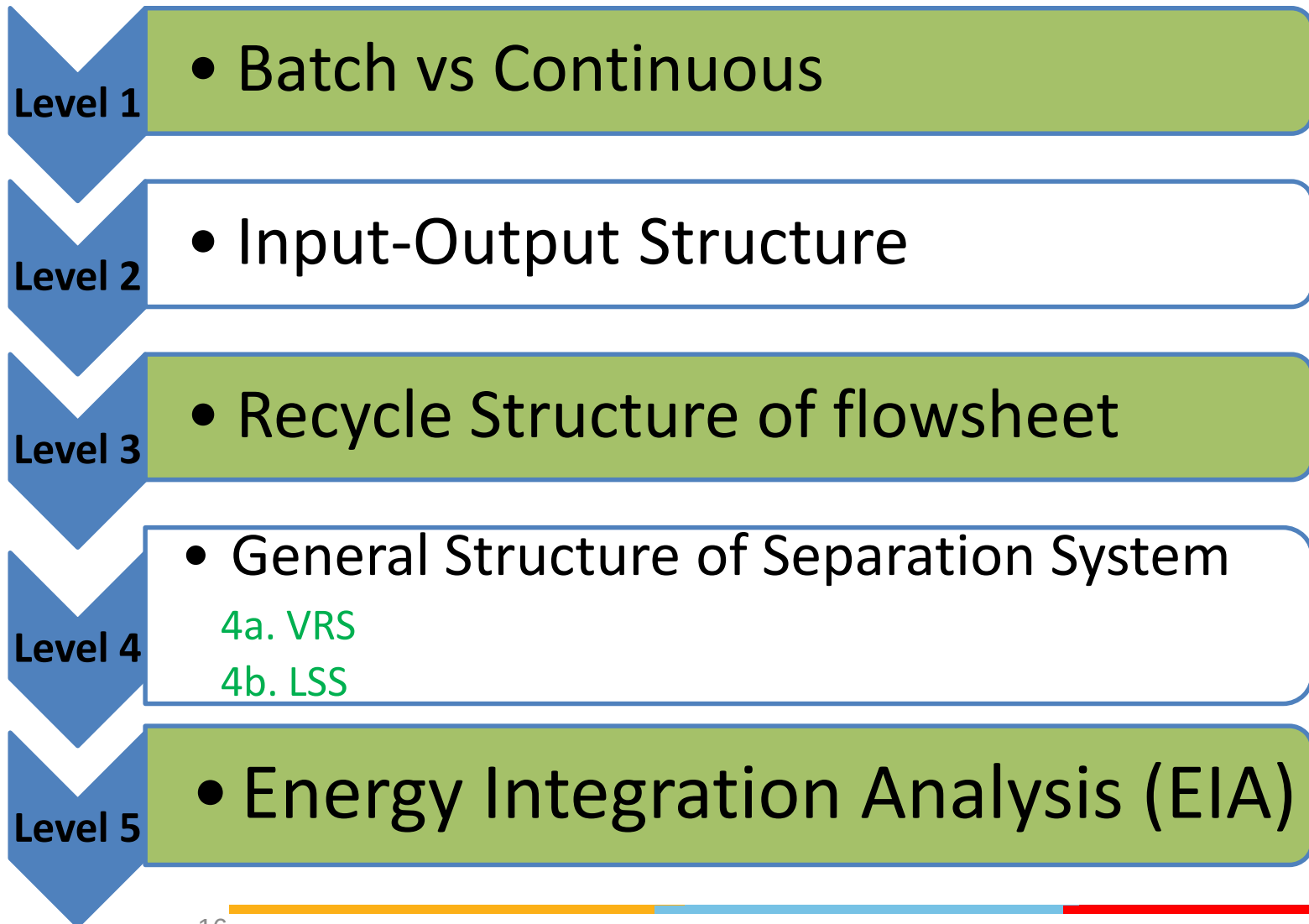
Need of Process Design Principles



Because of the under defined and open-ended nature of design problems, and because of the lower success rates, it is useful to develop a strategy for solving design problems



Hierarchy of Decisions



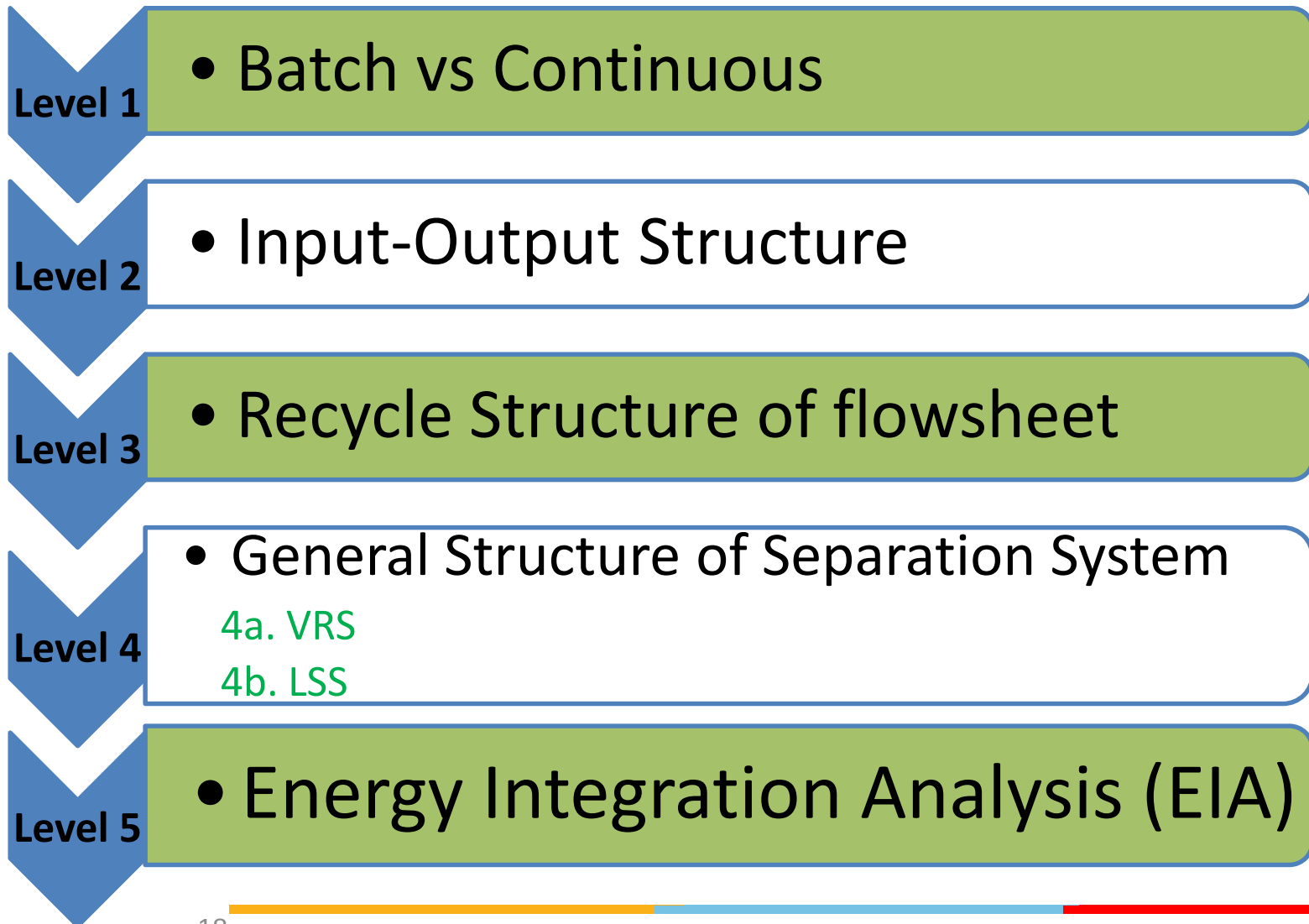
Input Information and Batch vs Continuous



Decision on Operating mode

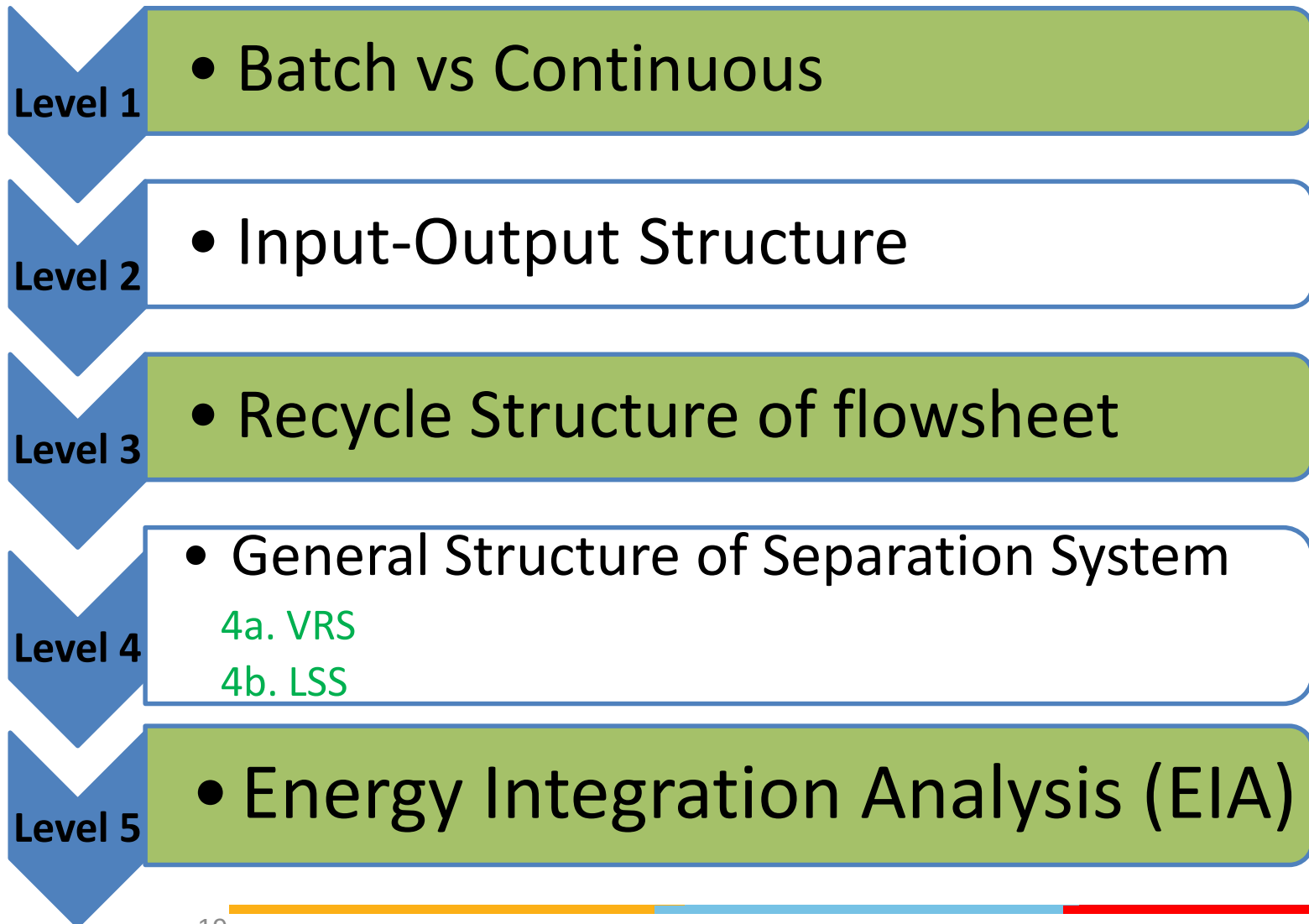


Hierarchy of Decisions



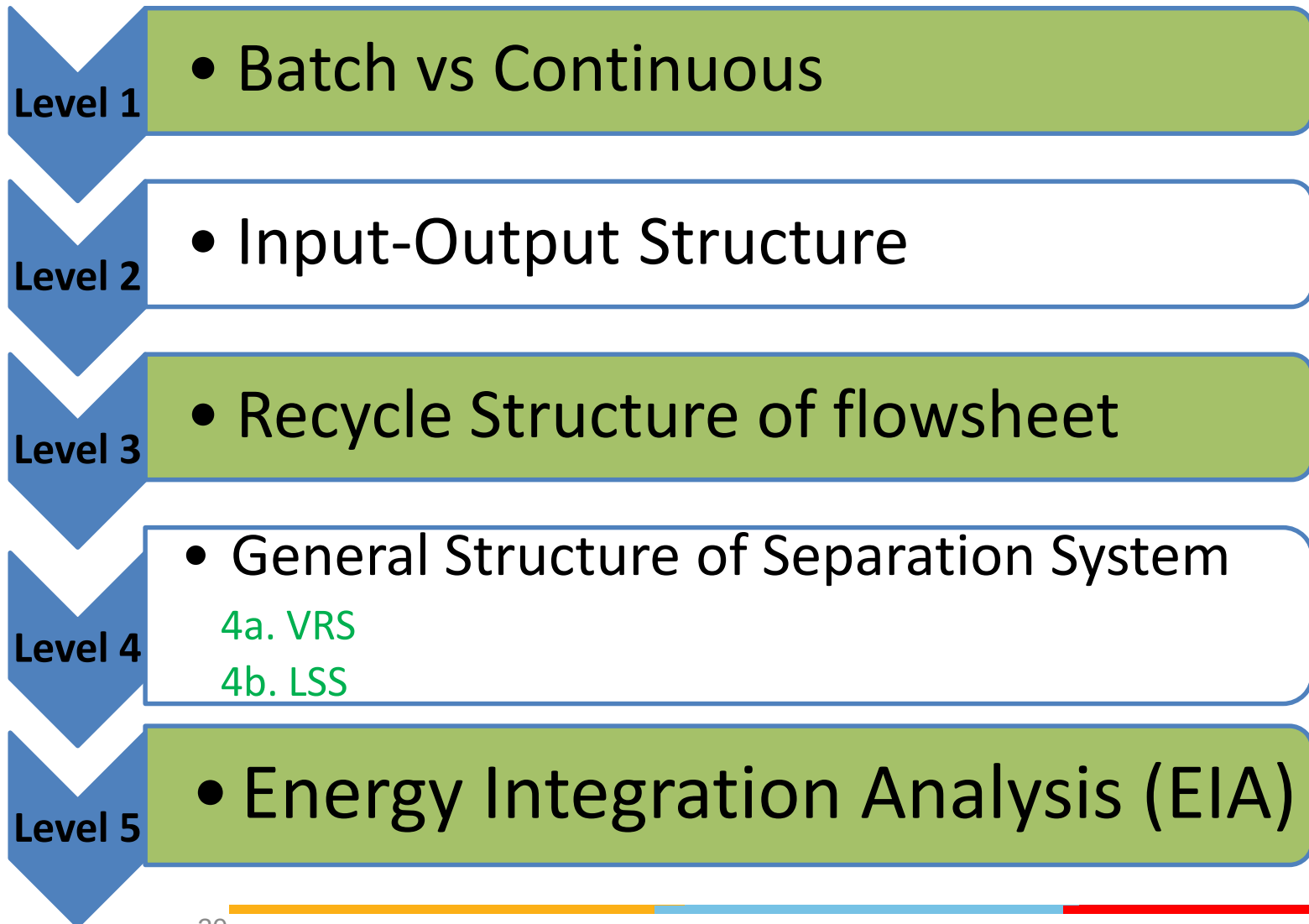


Hierarchy of Decisions



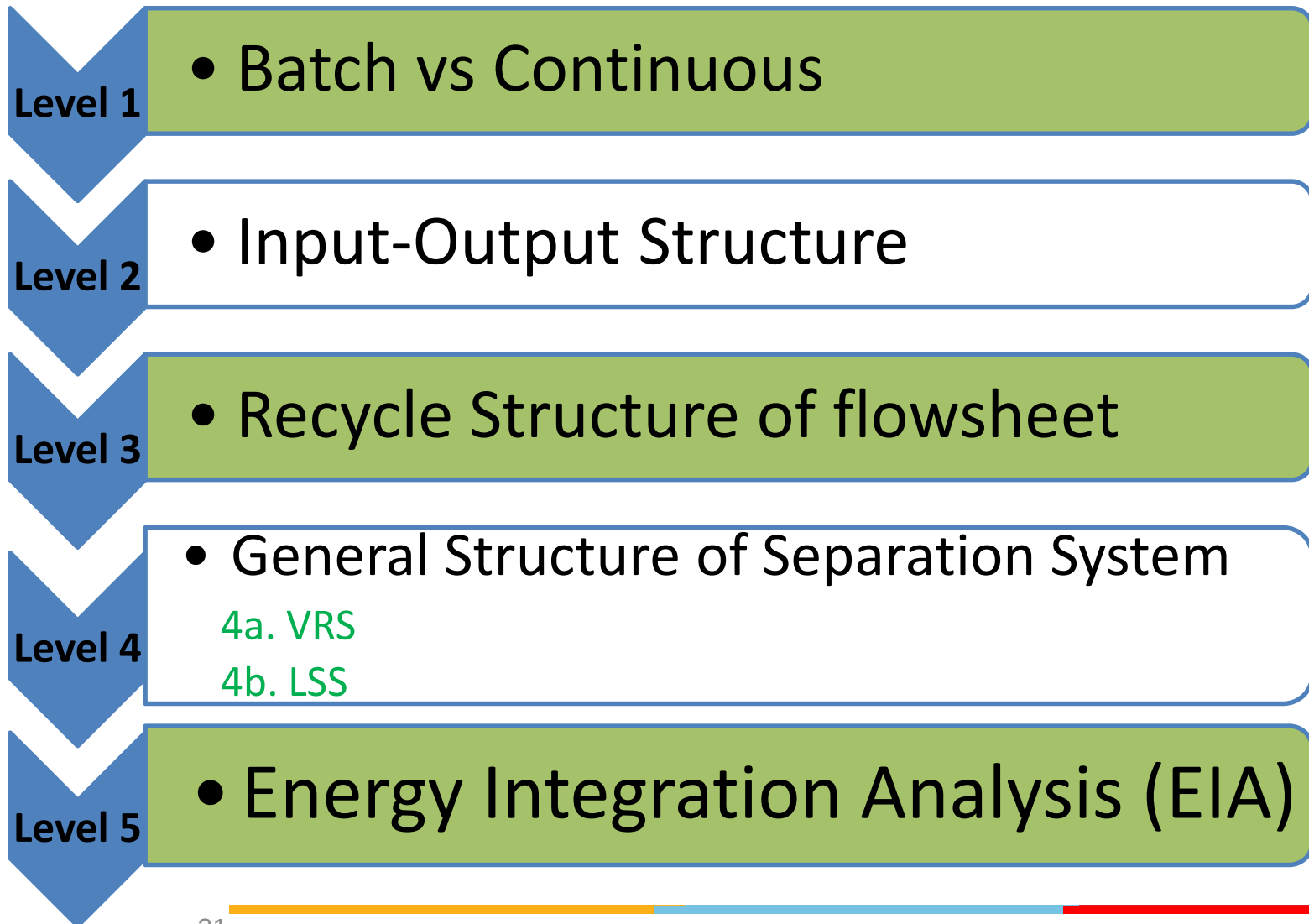


Hierarchy of Decisions





Hierarchy of Decisions





Thank You